

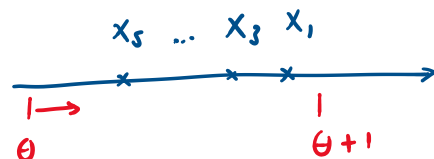
$\hat{\theta}_n$

Example: $X_1, \dots, X_n \stackrel{iid}{\sim} U(\theta, \theta+1)$.

Goal is to make an estimate of θ .

find m (or minimax lower bound)

$$\text{Likelihood} = \begin{cases} 1 & \text{if } \forall i \quad \theta \leq X_i \leq \theta+1 \\ 0 & \text{otherwise} \end{cases}$$

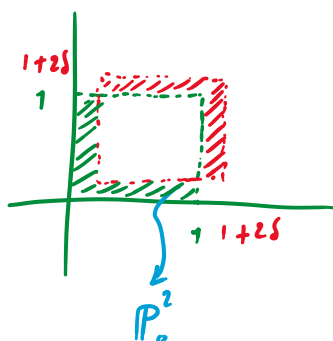


$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E} \{ \Phi(\rho(\hat{\theta}, \theta)) \} \geq ? \quad \{0, 2\delta\}$$

$$m \geq \frac{1}{2} \Phi(\delta) [1 - \|\underline{P}_\theta^n - \underline{P}_{2\delta}^n\|_{TV}] \quad \begin{aligned} \Phi(x) &= x^2 \\ \rho &= \|\cdot - \cdot\|_2 \end{aligned}$$

$n=2$
 $U^2(0,1)$

$U^2(2\delta, 1+2\delta)$



$$\|P - q\|_{TV} = \frac{1}{2} \int |P(x) - q(x)| dx$$

= Area of shaded Green region

$$= 1 - (1-2\delta)^2$$

$$\text{in gen.} = 1 - (1-2\delta)^n$$

$$m \geq \underbrace{\frac{\delta^2}{2} [1 - (1-2\delta)^n]}_{g(\delta)}$$

$$g'(\delta) = \delta(1-2\delta)^n - n\delta^2(1-2\delta)^{n-1} = 0 \quad \delta = 0$$

$$1-2\delta - n\delta = 0 \Rightarrow \delta = \frac{1}{n+2}$$

$$m \geq \frac{1}{2} \frac{1}{(n+2)^2} \left[1 - \frac{2}{n+2} \right]^n = \omega\left(\frac{1}{n^2}\right)$$

$\underbrace{\left[1 - \frac{2}{n+2} \right]^n}_{> 1/3} \xrightarrow{n \rightarrow \infty} e^{-2}$

□

Decomposable Distribution Distance

$$D_{KL}(P \parallel q) = \int p(x) \log \frac{p(x)}{q(x)} dx$$

$$D_{KL}(P^n \parallel q^n) = n D_{KL}(P \parallel q).$$

$$= \int p(x_1) p(x_2) \dots p(x_n) \log \frac{p(x_1) \dots p(x_n)}{q(x_1) \dots q(x_n)} dx_1 \dots dx_n$$

$$= \int p(x_n) \dots p(x_2) p(x_1) \left[\sum_i \log \frac{p(x_i)}{q(x_i)} \right] dx_1 \dots dx_n$$

$$= \sum_i \int p(x_n) \dots p(x_1) \log \frac{p(x_i)}{q(x_i)} dx_1 \dots dx_n$$

$$= \sum_i D_{KL}(P \parallel q) = n D_{KL}(P \parallel q)$$

Q: What is the relationship of the KL div & TV dist?

Pinsker's Ineq: $\|P - q\|_{TV} \leq \sqrt{\frac{1}{2} D_{KL}(P \parallel q)}$

$$n \geq \frac{\Phi(\delta)}{\frac{\delta}{2}} \left[1 - \sqrt{\frac{1}{2} D_{KL}(P_0 \parallel P_{2\delta})} \right]$$

Proof: $\|P - q\|_{TV} = \sup_A |P(A) - Q(A)|$

$$\|P - q\|_{TV} = P(A^+) - Q(A^+).$$

$$a := Q(A^+)$$

$$\delta := P(A^+) - Q(A^+)$$

$$a + \delta = P(A^+)$$

$$P(\bar{A}^*) = 1 - a - \delta$$

$$Q(\bar{A}^*) = 1 - a$$

Variational Bound on D_{KL}

$$D_{KL}(P \parallel q) \geq \mathbb{E}_P\{f\} - \log \mathbb{E}_q\{e^f\}$$

for any measurable f .

$$\begin{aligned} \text{sub proof: } \log \mathbb{E}_q\{e^f\} &= \log \mathbb{E}_P\left\{\frac{q}{P} \cdot e^f\right\} \stackrel{(\text{Jensen})}{\geq} \mathbb{E}_P\left\{\log\left(\frac{q}{P} e^f\right)\right\} \\ &= \mathbb{E}_P\left\{\log \frac{q}{P} + f\right\} = \underbrace{\mathbb{E}_P\left\{\log \frac{q}{P}\right\}}_{-D_{KL}(P \parallel q)} + \mathbb{E}_P\{f\} \end{aligned}$$

$$\Rightarrow D_{KL}(P \parallel q) \geq \underbrace{\mathbb{E}_P\{f\}}_{(1)} - \underbrace{\log \mathbb{E}_q\{e^f\}}_{(2)} \quad \square$$

$$f_{(x)} = \begin{cases} u & x \in A^* \\ v & x \notin A^* \end{cases}$$

$$\textcircled{1} \quad \mathbb{E}_P\{f\} = P(A^*)u + P(\bar{A}^*)v = (a+\delta)u + (1-a-\delta)v$$

$$\begin{aligned} \textcircled{2} \quad \log \mathbb{E}_q\{e^f\} &= \log\{Q(A^*) \cdot e^u + Q(\bar{A}^*) \cdot e^v\} \\ &= \log\{ae^u + (1-a)e^v\} \end{aligned}$$

$$D_{KL}(P \parallel q) \geq \underbrace{(a+\delta)u + (1-a-\delta)v - \log\{ae^u + (1-a)e^v\}}_{g(u,v)}$$

$$\frac{\partial g}{\partial u} = 0 = (a+\delta) - \frac{ae^u}{ae^u + (1-a)e^v} = 0$$

$$\frac{\partial g}{\partial v} = 0 = (1-a-\delta) - \frac{(1-a)e^v}{ae^u + (1-a)e^v} = 0$$

$$(a+\delta) [ax + (1-a)y] = ax$$

$$(a+\delta)ax - ax = (a+\delta)(a-1)y$$

$$a((a+\delta) - 1)x = (a-1)(a+\delta)y$$

$$\underbrace{a}_{\leftarrow} \underbrace{(1-a-\delta)}_{\rightarrow} x = \underbrace{(1-a)(a+\delta)}_{\leftarrow} y$$

$$\left(\frac{a}{a+\delta}\right)^x = \frac{(1-a)}{1-a-\delta} y$$

$$x = \frac{a+\delta}{a}, \quad y = \frac{1-a-\delta}{1-a}$$

$$u = \ln\left(\frac{a+\delta}{a}\right), \quad v = \ln\left(\frac{1-a-\delta}{1-a}\right)$$

$$D_{KL}(P \parallel Q) \geq (a+\delta) \ln\left(\frac{a+\delta}{a}\right) + (1-a-\delta) \ln\left(\frac{1-a-\delta}{1-a}\right)$$

$$- \ln\left(\frac{a}{a+\delta} + (1-a)\left(\frac{1-a-\delta}{1-a}\right)\right)$$

$$= D_{KL}\left(\begin{bmatrix} a+\delta \\ 1-a-\delta \end{bmatrix} \parallel \begin{bmatrix} a \\ 1-a \end{bmatrix}\right) \geq 2\delta^2$$

$\|P-Q\|_{TV}$

$$g(\delta) = (a+\delta) \ln\left(\frac{a+\delta}{a}\right) + (1-a-\delta) \ln\left(\frac{1-a-\delta}{1-a}\right) - 2\delta^2 \geq 0$$

$$1 \geq \delta \geq 0$$

$$g'(\delta) = \ln\left(\frac{a+\delta}{a}\right) + 1 - \ln\left(\frac{1-a-\delta}{1-a}\right) - 4\delta$$

$$g'(0) = 0$$

$$g''(\delta) = \frac{1}{a+\delta} + \frac{1}{1-a-\delta} - 4$$



$$g''(0) = \frac{1}{a} + \frac{1}{1-a} - 4 = \frac{1}{a(1-a)} - 4 > 0$$

$\forall s \quad g''(s) \geq 0 \Rightarrow g(0)$ is the min. amount of g

$$D_{KL}(P \parallel q) \geq D_{KL}(B(a+s) \parallel B(a)) \geq 2s^2$$

$$\underbrace{\|P - q\|_{TV}}_{\delta} \leq \sqrt{\frac{1}{2} D_{KL}(P \parallel q)} \quad \square$$